C-B-816G
August 9, 1977
SUPERSEDING
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August 6, 1970

FEDERAL SPECIFICATION

BUTTERMILK, FLUID AND MILK, WHOLE, FRESH, CULTURED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1. Scope. This specification covers the requirements for pasteurized cultured buttermilk, cultured lowfat milk and cultured milk for use by all Federal agencies.

1.2 Classification.

1.2.1 Types. The products covered by this specification shall be of the following types as specified (see 6.1):

Type I - Cultured buttermilk (less than 0.50 percent milkfat).
Type II - Cultured lowfat milk (not less than 0.50 percent nor more than 2.0 percent milkfat).
Type III - Cultured milk (minimum 3.25 percent milkfat).

1.2.1.1 Types I and II shall contain added vitamins A and D (see 3.3.2), when specified by the procuring agency (see 6.1). When vitamins A and D are not specified or not specifically excluded by the procuring agency, they may be added.

1.2.1.2 For civil agencies. The minimum composition requirements for the finished products shall be governed by the local or State laws or regulations in the area where the finished product is delivered, but shall be not less than the composition specified in 3.3 for each type.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.)

FSC 8910

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Military Standard:

MIL-STD-175 - Equipment and Methods for the Handling of Milk and Milk Products in Bulk Milk Dispensing Operations.

(Copies of specifications, standards and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

U.S. Department of Health, Education and Welfare

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

Grade "A" Pasteurized Milk Ordinance - 1965 Recommendations of the U.S. Public Health Service, Public Health Service Publication No. 229

Grade A Condensed and Dry Milk Products - A Recommended Sanitation Ordinance for Condensed and Dry Milk Products Used in Grade A Pasteurized Milk Products

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

Sanitation Compliance and Enforcement Ratings of Interstate Milk Shippers - U.S.P.H.S.

(Application for copies should be addressed to the Food and Drug Administration Bureau of Foods, HFF-415, 200 C Street, Washington, DC 20204.)

U.S. Department of Agriculture

Brucellosis Eradication - Recommended Uniform Methods and Rules, Animal and Plant Health Inspection Service, U.S. Department of Agriculture Publication 91-1

Uniform Methods and Rules - Bovine Tuberculosis Eradication

(Application for copies should be addressed to the Animal and Plant Health Inspection Service, Veterinary Services, Federal Center Building, Hyattsville, MD 20782.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, P.O. Box 540, Benjamin Franklin Station, Washington, DC 20044.)

American Public Health Association

Standard Methods for the Examination of Dairy Products

(Application for copies should be addressed to the American Public Health Association, 1015 Eighteenth Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Association of Vitamin Chemists

Methods of Vitamin Assay

(Application for copies should be addressed to Association of Vitamin Chemists, 8946 Major Avenue, Morton Grove, IL 60653.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Materials.

3.1.1 Milk. The raw milk used in the preparation of this product shall be drawn from cows in herds accredited as tuberculosis-free and certified brucellosis-free by the U.S. Department of Agriculture, or herds that have passed an annual tuberculosis test and meet USDA requirements for an individually certified herd, or from cows in herds located in:

(1) A Modified Accredited Tuberculosis Area in accordance with Uniform Methods and Rules - Bovine Tuberculosis Eradication; and
(2) Either (a) A Certified Brucellosis Area; or
(b) Modified Certified Brucellosis Area; or
(3) An area in the process of being certified by the USDA in accordance with USDA Animal and Plant Health Inspection Service, Publication 91-1. The milk shall be practically free from colostrum, shall be fresh, wholesome, and normal in appearance and odor and shall be subject to inspection by the procuring agency or duly authorized representative.

3.1.1.1 Bacterial requirements. The raw milk shall not exceed 100,000/ml prior to commingling or 300,000/ml prior to pasteurization.

3.1.1.2 Source. The raw milk shall be produced in localities having and enforcing milk ordinances that conform to the requirements for Grade "A" raw milk for pasteurization as defined in the Grade "A" Pasteurized Milk Ordinance - 1965 Recommendations of the U.S. Public Health Service, and the production and processing facilities, and regulatory laboratories shall meet equivalent requirements of the Grade "A" Pasteurized Milk Ordinance - 1965 Edition. Only such raw milk shall be used as the source of milk solids if milk solids are added or reconstituted. Dry milk products shall conform to the requirements of the Grade A Condensed and Dry Milk Products - A Recommended Sanitation Ordinance for Condensed and Dry Milk Products Used in Grade A Pasteurized Milk Products.

3.1.2 Optional ingredients as used by supplier.

3.1.2.1 Citric acid, or its equivalent as sodium citrate, of food grade quality may be added to the extent of not more than 0.2 percent of the weight of the milk to aid in the production of diacetyl.

3.1.2.2 Salt. Salt to enhance flavor may be added. The salt shall be white, refined sodium chloride of food grade quality, free from extraneous material.

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3.1.2.3 Starter distillate. Starter distillate and other diacetyl containing flavoring materials may be added to the product.

3.1.2.4 Stabilizers and emulsifiers. Safe and suitable stabilizers and emulsifiers permitted under the provisions of 3.7 may be used for the purpose of improving body and texture and to prevent syneresis (wheying off or separation) of the product.

3.2 Preparation.

3.2.1 Heat treatment (all types). The fluid milk product shall be given a special high heat, long time heat treatment preparatory to cooling (see 3.2.2) and adding culture. A temperature of 185°F (85°C) for 30 minutes shall be considered minimal heat treatment. If previously pasteurized milk or skim milk is used, such product shall be reheated and given the same exposure as an unheated one. Cooling to 70° to 72°F (21° to 22°C) shall follow immediately after heat treatment and just prior to inoculation with lactic culture (see 3.2.3). Thermal treatment yielding equivalent results shall be considered acceptable, provided minimal pasteurization temperatures and time as prescribed in the Pasteurized Milk Ordnance are followed.

3.2.2 Cooling of the cultured product. Adequate means shall be provided for cooling the product promptly following the attainment of the proper degree of acidity (known as the breaking point). The equipment designed for proper cooling, in order to restrict and control excessive acid development, shall be such as to provide for uniform, gentle agitation and still break up the coagulum or lumps. Storing of the product shall be at 40°F (4°C).

3.2.3 Inoculation and incubation. The fluid product shall be inoculated with a sufficient quantity of desirable active commercial lactic acid culture to produce a smooth, uniform coagulum coupled with a characteristic, clean, pleasant, diacetyl flavor and aroma. Both the amount of culture (or bulk starter as the case may be) and the incubation time at 70° to 72°F (21° to 22°C) shall be based on good commercial practice so as to result in an end product meeting the requirements of 3.3.

3.2.4 Vitamin A and D. When vitamins A and D are specified, they shall meet the potency requirements of 3.3.2. Vitamin A content may be increased by the addition of a concentrate of Vitamin A dissolved in an appropriate carrier. Vitamin D content may be increased by the addition or concentrate of Vitamin D of food grade quality thereof in a food oil, but if such oil is not milkfat, the quantity thereof added shall be not more than 0.01 percent, by weight, of the finished product.

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3.3 Finished product.

3.3.1 Physical requirements. The finished products shall possess a pleasing and desirable characteristic aroma and acid flavor and shall be free from undesirable flavors and odors. (NOTE: Frequent flavor defects in commercial, cultured buttermilks are a "high acid" or over-ripened, coarse flavor or a "low acid" or "green" flavor, both of which are considered undesirable). The body shall be smooth, uniform, and of medium consistency. The products shall be free from gas resulting from abnormal fermentation, wheying off, and practically free from entrapped air. The product may, at the option of the purchaser (see 6.1), contain small flakes, pellets, or granules of milkfat (butter) distributed uniformly throughout the body of the product.

3.3.2 Analytical requirements.

Milkfat (percent)	Type I	Type II	Type III
(not more than)	0.50	2.00	
(not less than)		0.50	3.25
Milk solids-not-fat (percent) (not less than)	9.00	9.00	8.25
Phosphatase activity (Scharer Rapid Method) 1/	1 mcg.	1 mcg.	1 mcg.
Titratable acidity (as lactic acid)			
Percent	0.80 - 0.95	0.80 - 0.95	0.80 - 0.95
pH	4.4 - 4.5	4.4 - 4.5	4.4 - 4.5

Vitamin D, when specified, (all types) 400 U S P units per quart (0.95 liter).
Vitamin A, when specified, (all types) 2,000 U S P units per quart (0.95 liter).

1/ All types - activity equivalent less than 1 microgram of phenol per ml. of product.

3.3.3 Bacteriological requirements. The bacterial limit for all types of finished product shall not exceed a coliform count of 10 per ml.

3.4 Age of product. The finished product shall be delivered within 96 hours after the date of packaging.

3.5 Delivery temperature. The internal temperature of the product covered by this specification at the time of delivery shall not exceed 45°F (7°C).

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3.6 Plant qualification. The products shall be processed in establishments meeting the requirements of the Grade "A" Pasteurized Milk Ordinance of the FDA or to those plants having a pasteurization plant compliance rating of 90 or more as certified by a state milk sanitation rating officer and listed in the document titled "Sanitation Compliance and Enforcement Rating of Interstate Milk Shippers".

3.6.1 Equipment and Workmanship. Equipment, operations, sanitary practices and personnel practices used in production and transportation of these products shall be approved by the inspection agency, as designated by the procuring agency, in accordance with 3.6.

3.7 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection records of the examination and tests shall be complete and made available to the Government as specified in the contract or order.

4.2 Plant qualification conditions. The product furnished under this document shall be unacceptable if not produced in plants complying with 3.6.

4.3 Inspection.

4.3.1 Sampling. Procedures for the collection and holding of samples; the selection and preparation of apparatus, media, and reagents; and the analytical procedures, incubation, reading, and reporting of results shall be in compliance with Standard Methods for the Examination of Dairy Products, and the Official Methods of Analysis of the Association of Official Analytical Chemists.

4.3.2 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards, unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

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4.3.2.1 Milk. Records of the results of the examination and testing of the milk supply shall be available for review by the procuring agency. If upon examination of the records it is determined that the processing plant has processed raw milk or milk products not in compliance with the requirements of 3.1.1, the finished product shall be rejected.

4.3.2.1.1 Animal sources. Inspection shall be made to determine that cows supplying the milk used in the preparation of this product are in herds designated by the U.S. Department of Agriculture as tuberculosis-free and from herds designated brucellosis-free or in the process of being so designated. Determination of herds status regarding tuberculosis and brucellosis shall be made by contacting the nearest Office of the USDA Animal and Plant Health Inspection Service. (Location of this office can be ascertained by consulting the publication entitled Directory of Meat and Poultry Inspection Program Establishments and Officials). Nonconformance to either of the above requirements shall be cause for rejection of the lot.

4.3.2.1.2 Ratings. Raw milk for pasteurization, receiving stations, and pasteurizing plants shall be rated in accordance with Methods of Making Sanitation Ratings of Milksheds. The pasteurized milk rating of the supplier shall be at least 90. A minimum rating of 90 shall be required for approval of individual receiving stations and pasteurizing plants.

4.3.2.2 Multi-use single-service containers. Records of the quarterly tests for compliance with the bacterial requirements of 5.1.1 shall be examined by the procuring agency or a duly authorized representative. Noncompliance with the above requirements shall be cause for rejection of the lot or the involved quantity of finished product made therefrom, as applicable.

4.3.2.3 Optional ingredients. Acceptance of optional ingredients used by the supplier for compliance with the requirement of 3.1.2, as applicable shall be based on the examination of the applicable labels, invoices, and similar documents. This examination shall be made on each new lot of the component received and not less than once during each contract. A statement of compliance shall be provided by the supplier that the optional nonmilk ingredients used conform to the requirements of 3.1.2.

4.3.3 In-process examination.

4.3.3.1 Equipment, procedures and controls. Examination shall be performed during processing to determine compliance with 3.6.1 pertaining to equipment, procedures and controls. Records shall be maintained for specific procedures of time and temperature controls covering product included in the purchase contract. Noncompliance with the requirements of 3.6.1 as determined by examination of records or actual examination, shall be cause for rejection of the involved finished product.

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4.3.3.2 Preparation and formulation. Examination shall be made during preparation of the product to determine compliance with: 3.2.1, 3.2.2 and 3.2.3, type requirements of the finished product (see 3.3), and with vitamin fortification requirements of 3.2.4 or optional ingredient requirement of 3.1.2, when applicable. Formulation records covering involved product shall be maintained and made available for review. Noncompliance with the above referenced requirements as determined by examination of records or by actual examination, shall be cause for rejection of the involved quantity of finished product.

4.3.3.3 Examination of finished product characteristics. Examination shall be made of the finished product to determine compliance with the pertinent requirements of 3.3.1, as concerns flavor, body, texture and other specified physical attributes. Frequency of this examination shall be established by the procuring agency and as specified in the contract or order (see 4.1.1 and 4.3.1). Noncompliance with the above referenced requirements as determined by examination of records or by actual examination shall be cause for rejection of the involved quantity of finished product.

4.3.3.4 Examination for age and temperature of product at time of delivery. Determination of compliance with requirements of 3.4 and 3.5 shall be ascertained from storage records and code or marked date of packaging. Noncompliance with requirements of 3.4 and 3.5 shall be cause for rejection of the lot.

4.3.4 Sampling for verification testing. (Not applicable to military agencies). Unless otherwise specified (see 6.1) the procuring agency or duly authorized representative will select verification samples to check supplier compliance with applicable finished product requirements (see 3.3) at not less than the following rate:

Initial or retest period	Sampling rate 1/
(If supplier has not had a contract in previous 90 days; or if three or more of the last five samples tested from supplier are unsatisfactory).	One sample from each delivery until three out of the last five are satisfactory and then sample at routine testing rate.
Routine test period	
(At least three of the last five samples tested from supplier shall be satisfactory).	One sample per month.

1/ The total volume needed for all test (see 4.4) could be as much as a half-gallon (1.9 liters); if fewer tests are required (see 6.1) a proportionally smaller volume will be needed. Total volume may be contained in any combination of container sizes. A representative aseptically drawn sample should be obtained from containers larger than 1/2 gallon (1.9 liters).

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4.3.4.1 Product supplied under the provisions of this specification to more than one receiver from the same production lot by a supplier, may, for the purposes of this paragraph, be sampled as one delivery. This does not preclude more intensive sampling by the receiver of product suspected of noncompliance.

4.4 Tests. Test procedures and controls which differ from those specified herein may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government contracting officer determines that such procedures and controls do not provide, as a minimum, such quality assurance, the supplier will use the test procedures set forth herein. In case of disputes as to test results, the test methods specified herein will govern.

4.4.1 Chemical analyses. Chemical analyses shall be made in accordance with the following methods from the Official Methods of Analysis of the Association of Official Analytical Chemists (AOAC). Standards Methods for Examination of Dairy Products (SMEDP), Method of Vitamin Assay, Association of Vitamin Chemists (MVAAVC), or in accordance with methods that give equivalent results.

Test	Method	Chapter & Section
Milkfat	Babcock or Roese Gottlieb 1/	(AOAC) Chapter: Dairy Products Section: Milk
Milk Solids, nonfat	Total Solids (method 1) (Subtract percent milkfat from percent total solids)	(AOAC) Chapter: Dairy Products Section: Milk
Phosphatase activity	Scharer Rapid Phosphatase Test	(SMEDP) Chapter: Phosphatase Methods to Determine Pasteurization
Acidity	Acidity	(AOAC) Chapter: Dairy Products Section: Milk 2/
Hydrogen-Ion Concentration (pH)	Electrometric	(AOAC) Chapter: Beverages: Malt Beverages and Brewing Materials Section: Beer
Vitamin D	Vitamin D in milk	(MVAAVC) Vitamins and other Nutrients Bioassay Method

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Test	Method	Chapter & Section
Vitamin A	Colorimetric (Carr-Price Blue Color)	(MVAAVC) Methods of Vitamin Assay

1/ For Military agencies the Roese-Gottlieb shall be the official method. For Civil agencies the Babcock test may be used, but in case of a dispute the Roese-Gottlieb method shall be final.

2/ Weigh sample.

4.4.2 Bacteriological examination. Bacteriological examination shall be made in accordance with the methods described in Standard Methods for the Examination of Dairy Products. The procedures shall be those specified therein for coliform test with solid media at 32°C.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level C. Product shall be delivered in single or multi-service containers of the capacity specified in the contract or order (see 6.1), or when specified (see 6.1), in sealed milk dispenser containers complying with MIL-STD-175.

5.1.1 All containers shall be sound and clean and shall afford adequate and proper protection of the contents from contamination. The residual bacterial count of multi-use and single-service containers used for packaging pasteurized milk and milk products, tested in accordance with 4.4.2, shall not exceed one per ml of capacity or 50 colonies per 8 square inches (5161.3 mm^2) of product-contact surface in three out of four samples taken at random on a given day.

5.1.2 All containers, except dispenser containers, shall be filled and closed by sanitary mechanical methods. Containers fabricated or formed at the supplier's plant shall be suitably protected from contamination during fabrication, forming and filling. The bulk milk dispenser containers shall be filled in a manner that will protect the top of the container effectively during filling, and after filling shall be immediately closed and sealed.

5.1.3 The pouring lip of all containers closed by cap or cover shall be completely covered to at least its largest diameter by the cap or cover. Paperboard milk containers which do not form a pouring lip until opened are exempt from this requirement.

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5.2 Packing, level C. Unless otherwise specified (see 6.1), the product shall be packed in containers in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Labeling and marking. Unless otherwise specified (see 6.1), any commercial labeling and marking complying with the local or state milk ordinance or regulations will be acceptable.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type product required (see 1.2).
(c) When vitamins A and D are required or may not be added (see 1.2.1.1).
(d) When types I and II product may contain small flakes, pellets, or granules of milkfat (butter) (see 3.3.1).
(e) When verification testing may be waived (see 4.3.4).
(f) Type and capacity of container required (see 5.1).
(g) When alternative level C packing is required (see 5.2).
(h) When other than commercial labeling is required (see 5.3).

6.2 Destination inspection (for civil agencies). When the finished product has been inspected and passed at point other than destination, the contract should require that the product be inspected at destination for condition and quantity only. Unless otherwise specified, such inspection should be made by the receiving facility. However, inspection for quality may be made by the procuring agency or a duly authorized representative, when deemed necessary to verify contract compliance.

6.3 Intended use and technical information. The products covered by this specification are intended as items of limited shelf life. Even though the products are already sour when delivered, they should nevertheless be considered as perishable and must be given equal refrigeration protection to that given to other perishable dairy products. The shelf life at 40°F (4°C) is approximately 7 days from time of production. These products may be used as a plain beverage or mixed with fruit juices or flavoring. They also may be used in baking such products as pancakes, waffles, cakes and cookies.

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Army - GL

Civil Agency Coordinating Activities:

AGR - AMS
GSA - FSS
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